

NASSAU CANDY DISTRIBUTOR

Product Line Profitability & Margin Performance Analysis

Business Analyst Internship Project • Unified Mentor

Business objective: Move beyond sales volume to understand which products and divisions truly drive profit, where margin risks exist, and how profitability is concentrated.

Project at a Glance

TRANSACTION RECORDS	UNIQUE ORDERS	PRODUCTS	DIVISIONS
10,194	8,549	15	3

Analytical Story

Raw Data → Data Cleaning → EDA → KPI Analysis → Product Profitability → Division Analysis → Pareto → Margin Risk → Recommendations → Dashboard

Core Questions

- Which products generate the most gross profit and margin value?
- How does profitability differ across product divisions?
- How concentrated is revenue and profit among a small number of products?
- Which products show cost-heavy or weak-margin characteristics?

Key principle: A strong Business Analyst does not stop at describing the data—the analysis should lead to a defensible business action.

1. Data Quality, EDA & KPI Framework

The first analytical step was to establish whether the dataset was fit for profitability analysis.

TOTAL SALES	GROSS PROFIT	GROSS MARGIN	UNITS SOLD
\$141,783.63	\$93,442.80	65.9%	38,654

Validation Performed

- Checked 10,194 records across 8,549 unique orders and 59 states/provinces.
- Reviewed Sales, Units, Gross Profit and Cost for missing values, zero-sales records and negative financial values.
- Standardized and analyzed product and division labels using the supplied fields.
- Used Order Date for time-based profitability analysis.

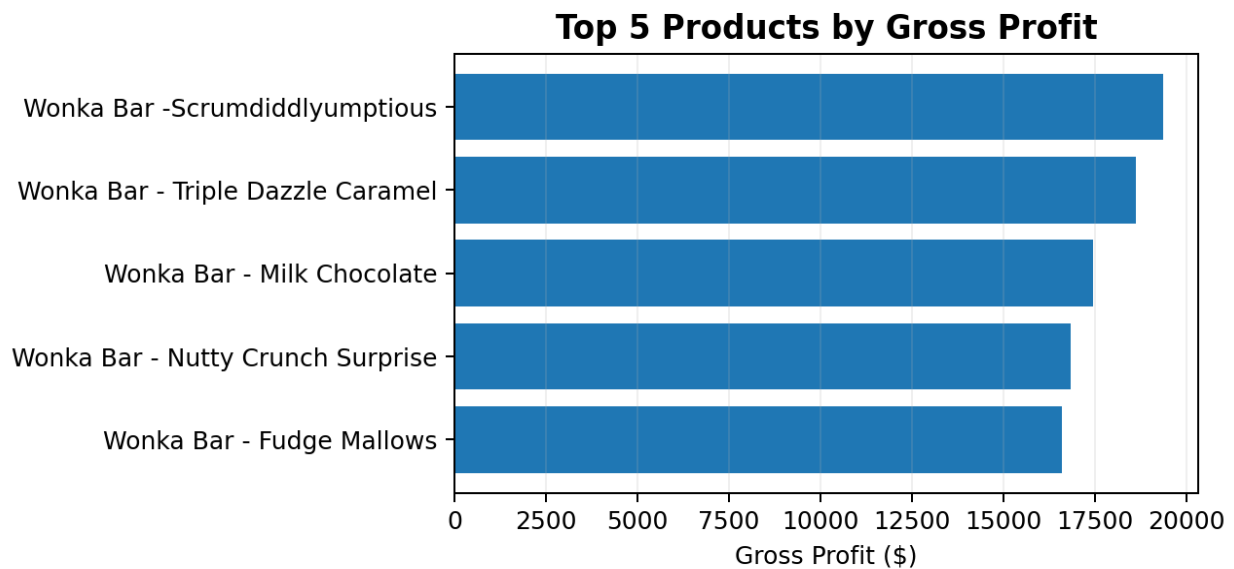
Data-quality finding: Order Date parses from 2024-01-02 to 2025-12-31, while Ship Date parses from 2026-06-30 to 2030-06-28. The Ship Date field should be validated against the original source before any logistics or shipment-timing conclusions are made.

KPI Framework

KPI	Formula	Business Use
Gross Margin	Gross Profit ÷ Sales	Measures profitability efficiency
Profit per Unit	Gross Profit ÷ Units	Shows value generated per unit
Revenue Contribution	Product Sales ÷ Total Sales	Shows revenue importance
Profit Contribution	Product Profit ÷ Total Profit	Shows profit importance
Cost-to-Sales	Cost ÷ Sales	Highlights cost intensity

2. Product & Division Profitability

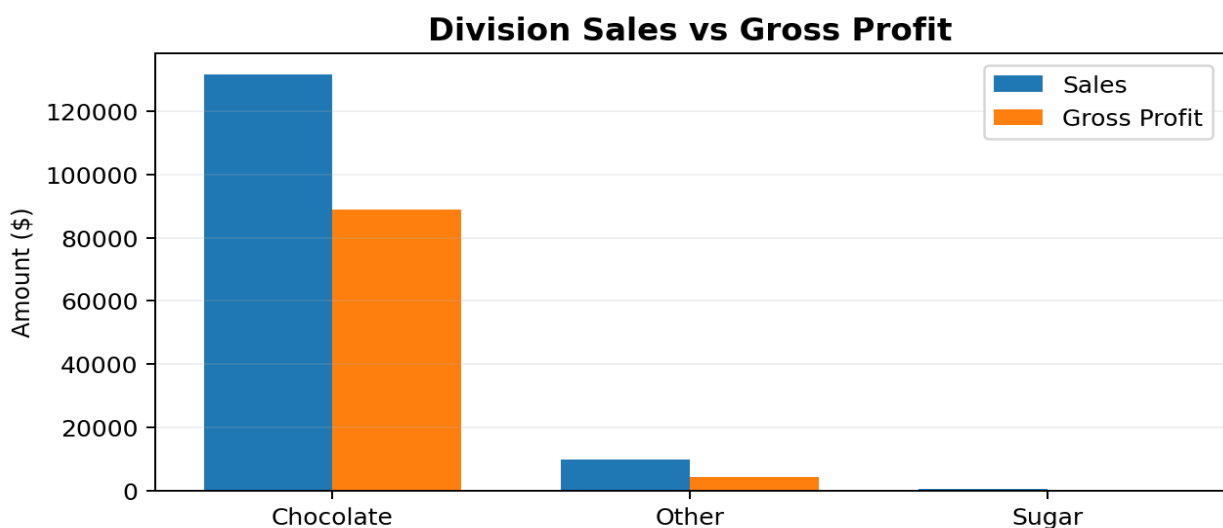
The analysis evaluates financial scale and profitability efficiency together.



Top Product Findings

- **Wonka Bar - Scrumdiddlyumptious** — Sales \$27,874.80; Gross Profit \$19,357.50; Gross Margin 69.4%.
- **Wonka Bar - Triple Dazzle Caramel** — Sales \$28,485.00; Gross Profit \$18,610.20; Gross Margin 65.3%.
- **Wonka Bar - Milk Chocolate** — Sales \$26,867.75; Gross Profit \$17,443.37; Gross Margin 64.9%.
- **Wonka Bar - Nutty Crunch Surprise** — Sales \$23,574.95; Gross Profit \$16,819.95; Gross Margin 71.3%.
- **Wonka Bar - Fudge Mallows** — Sales \$24,890.40; Gross Profit \$16,593.60; Gross Margin 66.7%.

Product insight: The top five products generate approximately 95.1% of total gross profit. This means product-level prioritization should focus on profit contribution, not sales alone.

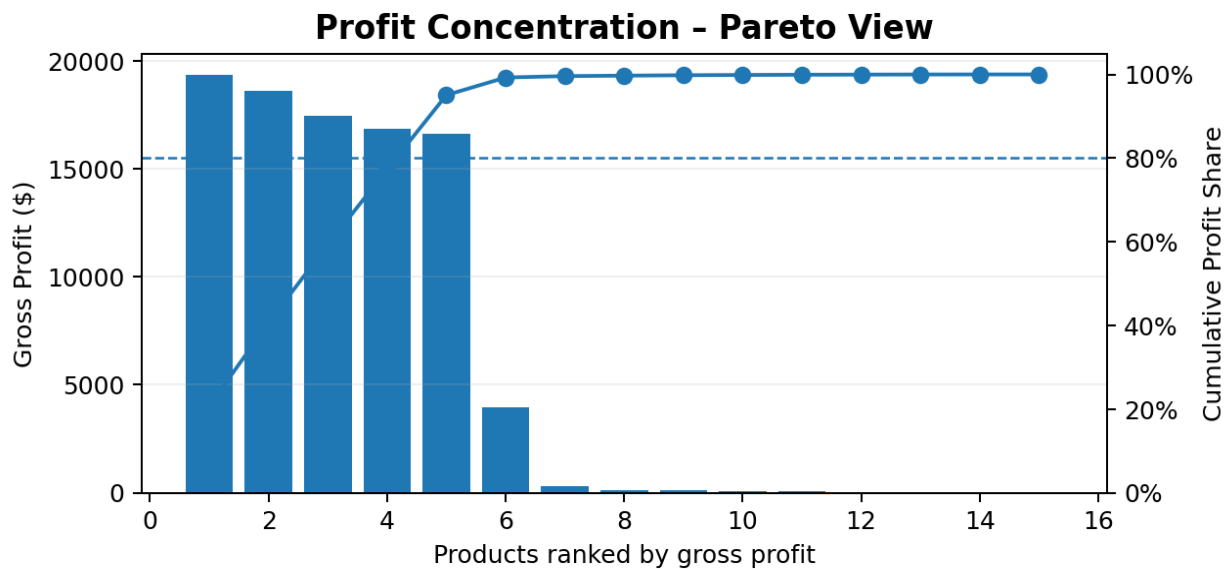


Division Findings

- **Chocolate** — Sales \$131,692.90; Gross Profit \$88,824.62; Gross Margin 67.4%; Profit Contribution 95.1%.
- **Other** — Sales \$9,663.25; Gross Profit \$4,333.45; Gross Margin 44.8%; Profit Contribution 4.6%.
- **Sugar** — Sales \$427.48; Gross Profit \$284.73; Gross Margin 66.6%; Profit Contribution 0.3%.

3. Pareto Concentration & Margin Risk

The portfolio is profitable, but the source of that profit is highly concentrated.



PRODUCTS TO EXCEED 80% PROFIT	TOP 5 PROFIT SHARE	OVERALL GROSS MARGIN	KAZOOKLES MARGIN
5	95.1%	65.9%	7.7%

Priority Margin-Risk Signal

Product	Sales	Cost-to-Sales	Gross Margin	Interpretation
Kazookles	\$1,205.75	92.3%	7.7%	Immediate investigation

Kazookles: approximately 92.3% of sales are consumed by cost, leaving only about 7.7% gross margin. Recommended sequence: pricing review → supplier/cost review → cost reduction → portfolio rationalization review if economics cannot improve.

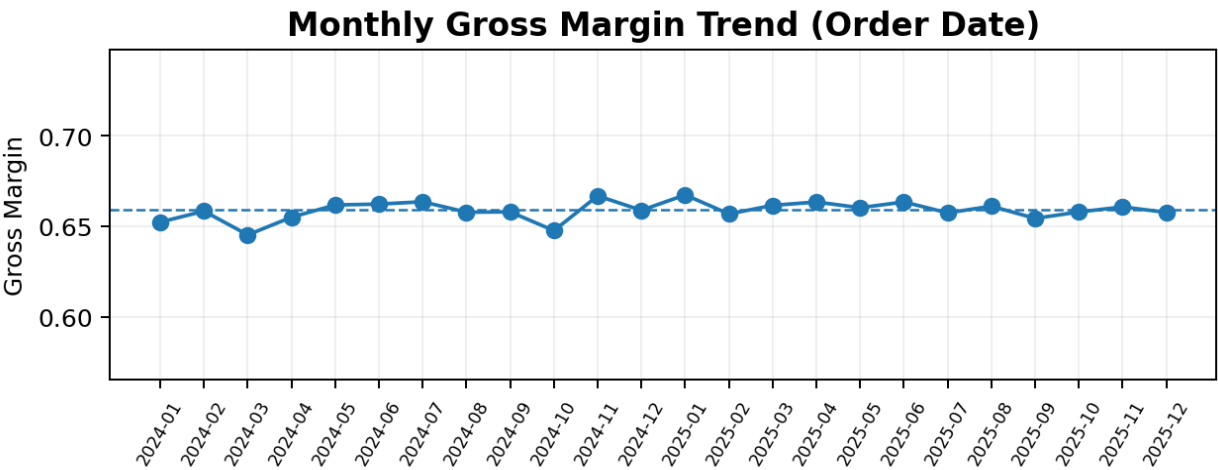
Important: A low-margin product should not automatically be discontinued. Risk should be prioritized using margin, sales scale and profit contribution together.

4. Business Recommendations & Decision Framework

The final step is to convert the analysis into practical management actions.

1 PROTECT	Protect the core profit-generating products because a small number of products contribute most of total profit.
2 INVESTIGATE	Review products with unusually high cost-to-sales ratios and structurally weak unit economics.
3 PRIORITIZE	Rank risks using Margin + Sales Scale + Profit Contribution rather than a single metric.
4 DIVERSIFY	Explore profitable portfolio expansion to reduce excessive dependence on a small product/division set.
5 MONITOR	Establish recurring monthly tracking of Sales, Gross Profit, Gross Margin, Cost-to-Sales and Product Contribution.
6 VALIDATE	Resolve the Ship Date source inconsistency before making logistics, delivery or route-efficiency conclusions.

Monthly Margin Context



Dashboard outcome: The Streamlit solution turns the analysis into interactive decision support with executive KPIs, product profitability, division performance, margin-risk flags, Pareto analysis, data-quality checks, date/division/margin filters and product search.

FINAL BUSINESS MESSAGE

Manage the portfolio using profit contribution, margin efficiency, cost intensity and concentration risk.